



SALA 2026 - USFQ - HACKATHON

GalapagOS: AI for Fossil-Free Galápagos

AI Model to Predict solar energy to reduce diesel dependency in San Cristóbal

- Jesús Zelaya (Perú)
- Jorge Torres (Ecuador)
- María Riojas (Perú)
- Brian Morales (Perú)
- Eduardo Jauregui (Perú)
- Michelle Jones (Costa Rica)
- Salvador Yábar (Perú)
- Adrian Acarapi (Bolivia)



**UNESCO WORLD NATURAL HERITAGE
1978**



Galápagos Energy Challenge

FOSSIL-FUEL ELECTRICITY GENERATION THREATENS THE GALÁPAGOS ECOSYSTEM

70%

of electricity in Galápagos
comes from fossil fuels

2nd

San Cristóbal has the
2nd highest electricity
demand

2040

Zero Fossil Fuel
target for Galápagos



THE PROBLEM:

There is no predictive system to decide when diesel generators should turn  or 

Diesel generators run **24/7**



High fossil fuel consumption



Unnecessary operational costs

DATA

4 Weather Stations

Coast → Summit transect (80m → 700m)

10 years of data

15-minute resolution

14 meteorological variables

Elevation gradient weather network



METHODOLOGY

1. The Predictor

XGBoost

predict solar ramp events

1-3 hours ahead

2. The Physics Engine

Convert solar radiation

$\text{W/m}^2 \rightarrow \text{kW generation}$

Renewable Dispatch Index

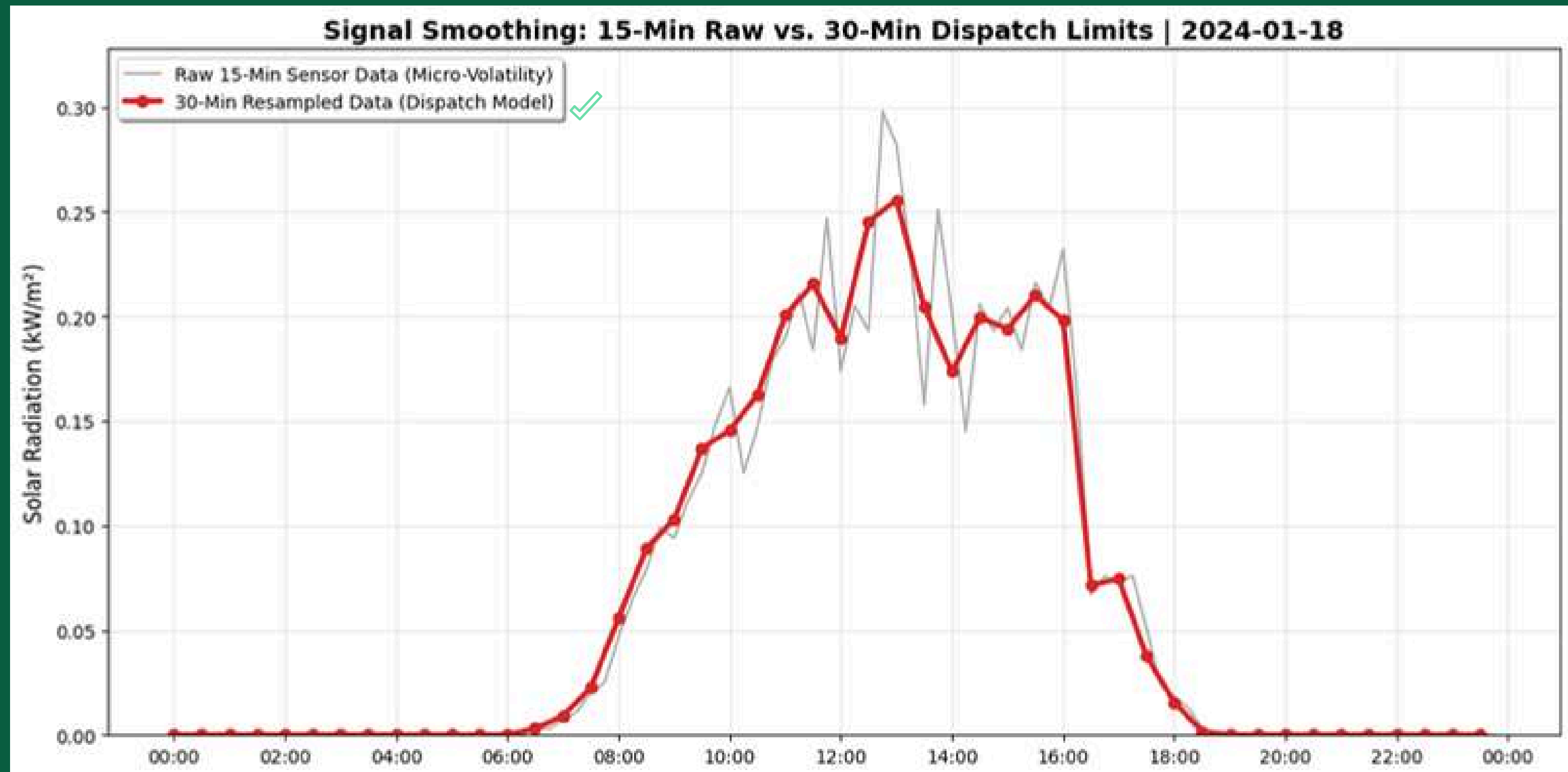
(green / yellow / red)

3. The Interface

Dashboard recommendation

"Power down Generator #2 at 14:00"

Due to the higher volatility observed in the 15-minute irradiance data, we chose to proceed with a 30-minute resolution for the training timesteps



MODEL CARD

POPULATION

Meteorological data from the Junco base since June 2015

TARGET

Radiation in 30 60, 90, 120, 150, 180 minutes

PERIODS

TRAINING PERIOD

2015/06 - 2024/01

150,087 total timesteps

TESTING PERIOD

2024/01 - 2026/03

37,522 total timesteps

METHODOLOGY

6 XGBoost Regression models for each target

FEATURES

59 features

Let's talk about Model Model Objectives



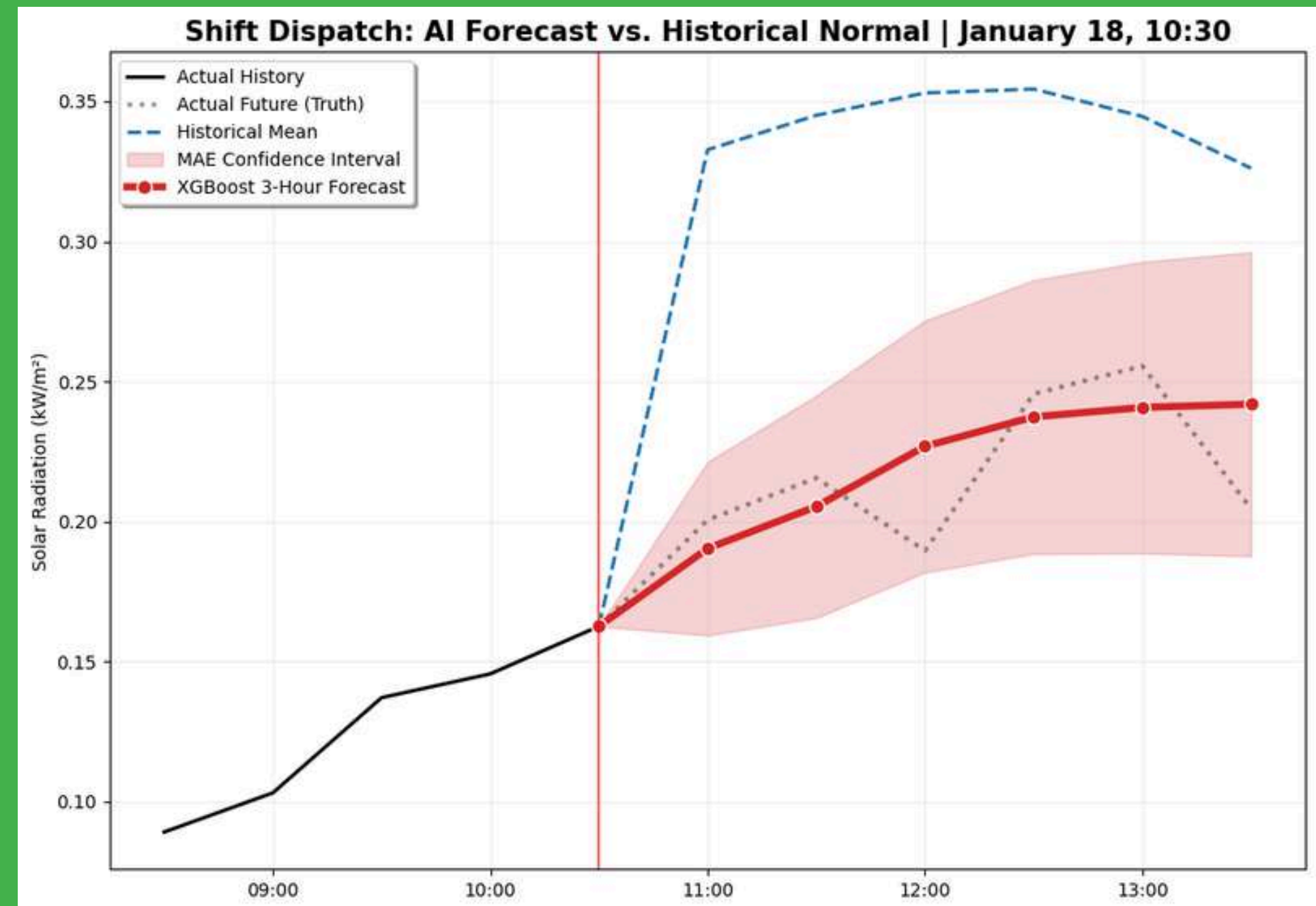
Dispatch Engine: 3-Hour Forecasting Window.



Intelligence: Uses local weather telemetry + historical memory to project 6 discrete steps of power availability.



Grid Result: Optimized Reliability. Provides the "Confidence Fan" needed to balance 1 MW of solar with diesel and battery reserves



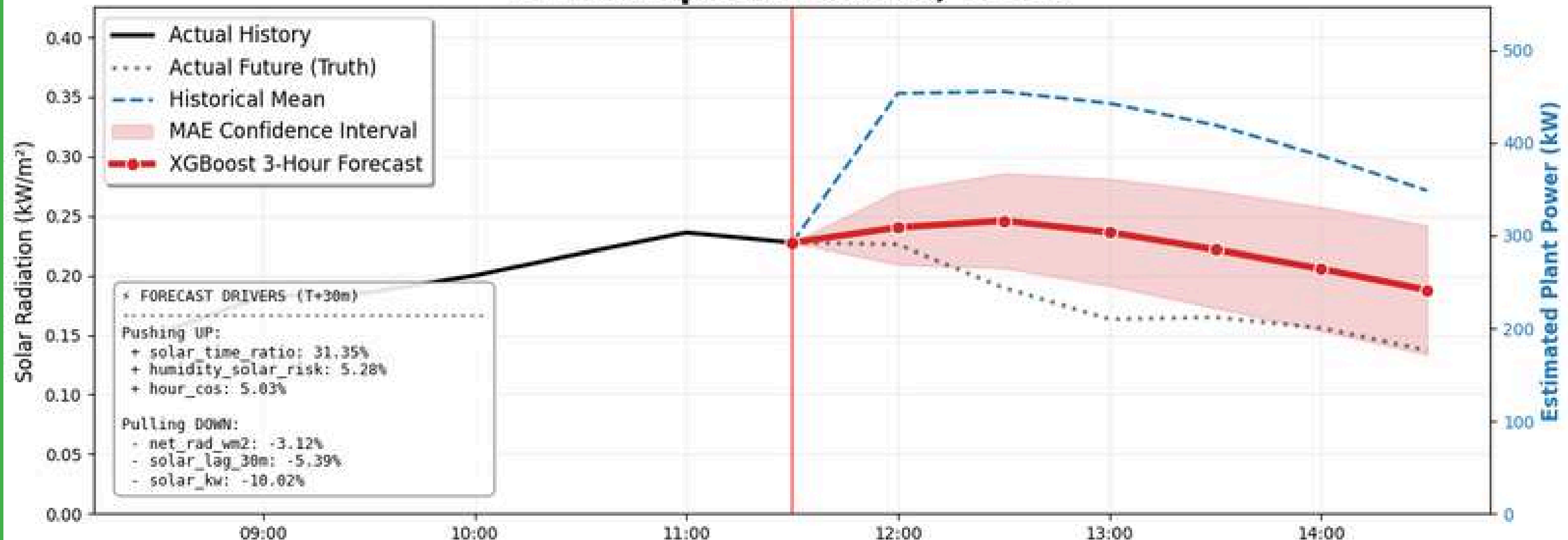
The model significantly outperforms the baseline, reducing the prediction error from 40% (temporal mean) to just 17% for highly accurate radiation forecasting

Top Features

Top Features:

- **solar_lag**: Provides historical data of the target.
- **solar_time_ratio**: Measures radiation intensity relative to the time of day.
- **hour_sin/cos**: Maps the 24-hour solar cycle to a circular coordinate system

Shift Dispatch: Oct 11, 11:30



THE OPERATIONAL GAP

Without prediction

- Diesel generators run 24/7 as "spinning reserve"
- ~\$500K per year wasted in fuel in San Cristóbal

With GalapaOS

- Predict solar generation for a 3-hour window
- Turn diesel generators ON/OFF at the right time

Predict the sun → Reduce diesel



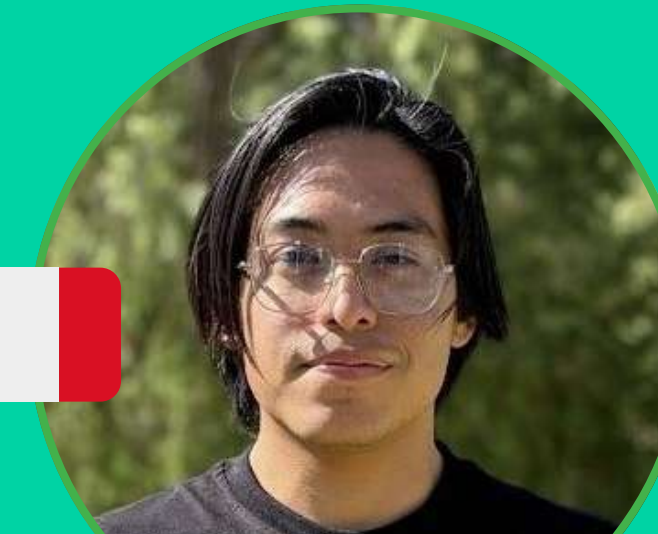
María Riojas
Cloud Dev | Political
Scientist | Firefighter



Eduardo Jáuregui
Data Scientist
Software Engineer



Jorge Torres
Telecom Engineer |
Cibersecurity Presales



Salvador Yábar - Electrical
Engineer | Renewable
Energy Researcher



Jesús Zelaya
Data Scientist
Mathematician



Michelle Jones Pérez
Human Systems Researcher
Anthropology | Geography



Brian Morales - Electronics
Engineer | Computer
Vision Researcher



Adrian Acarapi
Software Engineer I

Model Results

